Jeongwoo Shin

Updated: August 27, 2026

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Research Interest

Generative Modeling, Diffusion & Flow Matching, Reward-Finetuning of Diffusion Models, Stochastic Optimal Control, Diffusion/Flow Model Distillation, Long Video Generation, 3D-Consistent Video Generation.

Education

Graduate School of Data Science, Seoul National University (SNU)
Ph.D. Data Science *2024.03 – Current*
- Advisor: Joonseok Lee
- Research: Generative Modeling, Diffusion & Flow Matching
- GPA: 3.70 / 4.30

Graduate School of Data Science, Seoul National University (SNU)
M.S. Data Science *2022.03 – 2024.02*
- *Thesis: Self-Guided Masked Autoencoder*
- Advisor: Joonseok Lee
- GPA: 3.98 / 4.30

Seoul National University (SNU)
B.S. Landscape Architecture and Rural System Engineering *2016.03 – 2022.02*
- *Thesis: Adsorptive Removal of Methylene Blue from Aqueous Solution Using Carbon Nanospheres*
- GPA: 3.59 / 4.30

Publications

* *Equal contribution.* † *Corresponding author.*

[C4] Efficient Generative Modeling beyond Memoryless Diffusion via Adjoint Schrödinger Bridge Matching
Jeongwoo Shin, Jinhwan Sul, Joonseok Lee†, Jaewoong Choi†, Jaemoo Choi†
International Conference on Machine Learning (ICML) 2026

[C3] Equivariant Latent Alignment via Flow Matching under Group Symmetries
Sunghyun Kim*, Jaehoon Hahm*, Jeongwoo Shin, Joonseok Lee†
International Conference on Machine Learning (ICML) 2026

[C2] Latent Diffusion Models with Masked AutoEncoders
Junho Lee*, Jeongwoo Shin*, Hyungwook Choi, Joonseok Lee†
International Conference on Computer Vision (ICCV) 2025

[C1] Self-Guided Masked Autoencoder
Jeongwoo Shin, Inseo Lee, Junho Lee, Joonseok Lee†
Conference on Neural Information Processing Systems (NeurIPS) 2024

[P4] Efficient Adjoint Matching for Fine-tuning Diffusion Models
Jeongwoo Shin*, Dongsoo Shin*, Yuchen Zhu, Wei Guo, Yongxin Chen, Joonseok Lee†, Jaewoong Choi†, Jaemoo Choi†
Under Review at NeurIPS 2026

[P3] Diverse Motion Customization via Control-based Dynamic Optimization

Youngyoon Choi*, Kihyun Kim*, Jeongwoo Shin†, Joonseok Lee†
Under Review at NeurIPS 2026

[P2] No More Hit-and-Hope: Optimal Initial Noise Forging via Noise Shuffling
Suho Ryu, Jeongwoo Shin, Joonseok Lee†
Under Review at NeurIPS 2026

[P1] Optimal Crystal Flow for Fast and Reliable Crystal Generation
Seunghoon Yi, Youngwoo Cho, Jaehoon Hahm, Jeongwoo Shin, Soo Kyung Kim†, Jaegul Choo†, Joonseok Lee†, Hongkee Yoon†
Under Review at NeurIPS 2026

Awards & Scholarships

Research Fellowship for Ph.D. Candidates 2025.09 – 2026.08
Principal Investigator (PI) National Research Foundation of Korea (NRF)
- Project: *Optimal Latent Space for Effective Image Generation*
- Competitive individual research grant for Ph.D. students, funded by the Ministry of Education.
- Research on optimal latent spaces for image generation.
- Proposed a stochastic optimal control based generative process, resulting in [P4].

Research Experience

Data Refinement using Multi-Modal Foundation Models 2023.11 – 2024.09
Samsung Electronics Co., Ltd.
- Collaborative research project with Samsung Electronics.
- Worked on data refinement with large multi-modal foundation models for personalized search and generation.

Academic Service

Conference Reviewer
- ICML 2026 (*Silver Reviewer, Top 50%*), ECCV 2026, NeurIPS 2026

Teaching Experience

TA, SNU
M3239.005300: Machine Learning and Deep Learning I 2023 Spring

Languages

Programming: Python, C, C++
Deep Learning: PyTorch, TensorFlow, NumPy
Natural Languages: English (Fluent), Korean (Native)

References

Joonseok Lee 2022.03 - Current
M.S. & Ph.D. Advisor (SNU) joonseok@snu.ac.kr